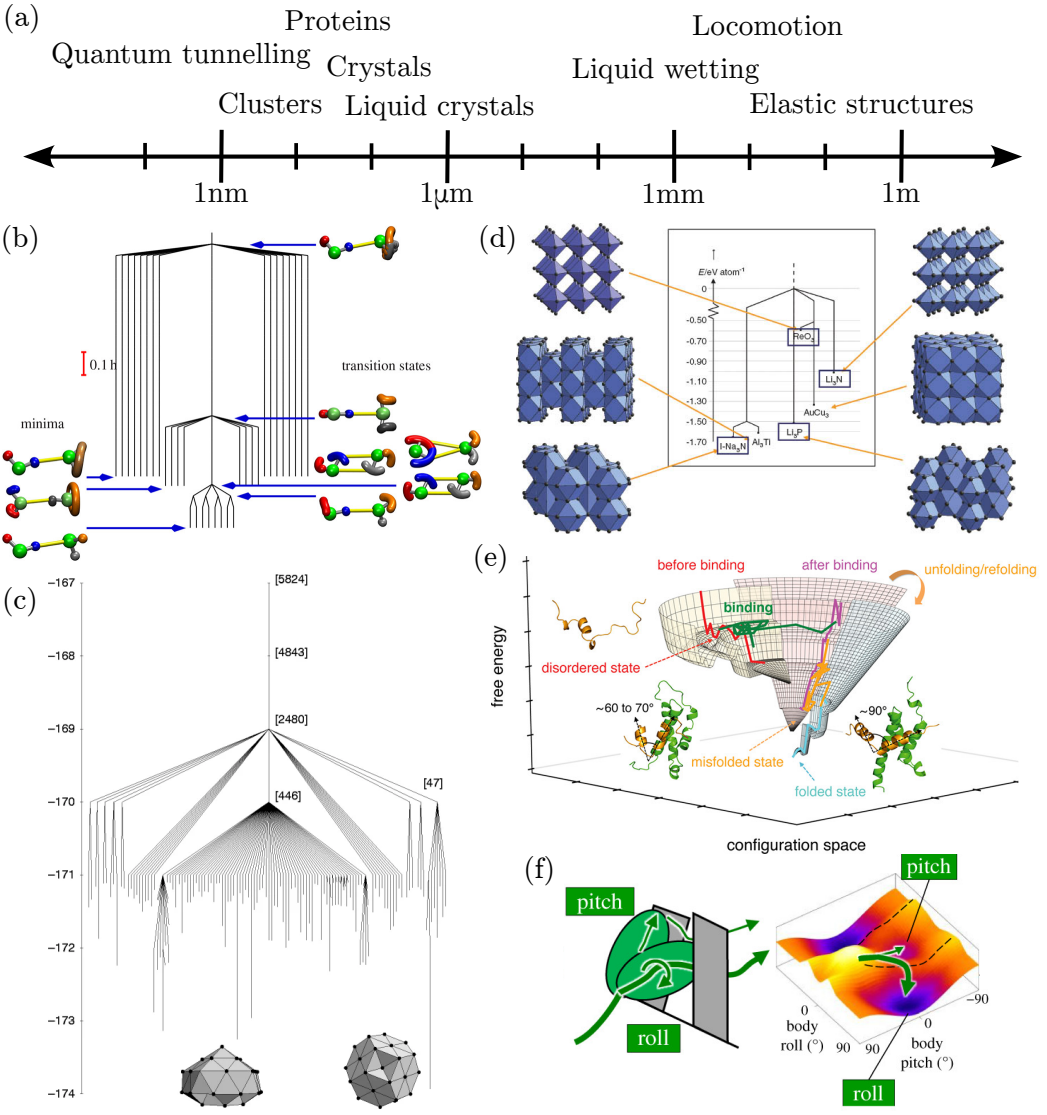

(a)
Proteins
Quantum tunnelling
Crystals
Locomotion
Liquid wetting
Clusters
Liquid crystals
Elastic structures
1nm
1μm
1mm
1m
(b)
0.1 h
transition states
minima
(c)
−167
−168
−169
−170
−171
−172
−173
−174
[5824]
[4843]
[2480]
[446]
[47]
(d)
E/eV atom⁻¹
0
-0.50
-0.70
-0.90
-1.10
-1.30
-1.50
-1.70
ReO3
Li3N
AuCu3
Li3P
I-Na3N
Al3Ti
(e)
before binding
after binding
unfolding/refolding
binding
free energy
disordered state
~60 to 70°
~90°
misfolded state
folded state
configuration space
(f)
pitch
roll
pitch
roll
0
body
roll (°)
90
90
0
body
pitch (°)
−90